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GAS TURBINE PARAMETER CORRECTIONS

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ABSTRACT

The various parameters appearing along an engine's gas path, such as flows, pressures, temperatures, speeds, etc., vary not only with power condition but also with the ambient conditions at the engine's inlet. Since a change in inlet temperature and/or pressure will contribute to an attendant change in a gas path parameter's value, it would be difficult to characterize the aero-thermodynamic relationships between gas turbine engine parameters, (even at a constant engine operating point) unless the ambient conditions are somehow accounted for. This is usually accomplished through the use of *corrected* engine parameters. Although most of these corrections are well known by practitioners in the industry, knowledge of their origin does not appear to be as commonplace. The purpose of this paper is to fill that gap and furnish a summary of the commonly used corrections for the "major" gas path parameters that are used in performance analysis, diagnostics and control design, and to offer a derivation of these corrections. We will suggest both an analytic approach as well as an empirical approach. The latter can be used to establish the correction for parameters not directly addressed in this paper, as well as to fine tune the correction factors when actual engine data is available.

m	Mass (lbm)
N	Speed (rpm)
P	Pressure (psia)
$\mathcal{P}$	Arbitrary gas path parameter
$\mathcal{P}^*$	Arbitrary gas path parameter corrected to standard day conditions
Q	Torque
R	Gas constant
S	Entropy
T	Temperature (degrees R)
$\dot{W}_a$	Air mass flow (lbs/sec)
$\dot{W}_f$	Fuel mass flow (lbs/sec)
θ	nondimensional temperature (ratio)
δ	nondimensional pressure (ratio)
ρ	Density
κ	Fluid thermal conductivity
v	Fluid velocity
μ	Fluid viscosity
η	Adiabatic efficiency
Γ	Flow capacity

NOMENCLATURE

A	Area
cp	Specific heat @ constant pressure (BTU/lbm °R)
cv	Specific heat @ constant volume (BTU/lbm °R)
D	Diameter
F_n	Net thrust (lbs)
g	Acceleration constant (32 ft/sec ²)
H	Heat transfer coefficient (BTU/hr ft ² °R)
HP	Horsepower
h	Enthalpy
I	Moment of inertia
J	Mechanical equivalent of heat (778 ft-lbs/BTU)
k	Ratio of specific heats (c_p/c_v)

Subscripts

1,2,3,...	Engine station number
s	Static quantity
t	Total quantity

Superscripts

*	Corrected quantity
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HISTORICAL PERSPECTIVE

The use of corrected parameters to describe the performance of a gas turbine appears to be as old as the machine itself [Warner (1945), Sanders (1946), Shepherd (1949)]. Earlier works [e.g. Capon, (1930)] reference many of the same parameters as applied to more general

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aero-thermodynamic machines such as air compressors. These typically take the form of so-called dimensionless quantities derivable from dimensional analysis, which has been the primary technique for establishing what I will refer to as the classical parameter corrections. Since this technique is somewhat old and is not necessarily known to all, the analytic derivations that are provided in this paper will use a more general method which relies only on a knowledge of basic calculus and a handful of thermodynamic principles. Since dimensional analysis has played such an important historical role in establishing these corrections, we will describe (briefly) its principle and derive a particular correction as an illustrative example. The following description follows that given by [Buckingham (1914)].

Let us assume that we have a physical relationship to be represented by n physical quantities $Q_1, Q_2, \dots, Q_n$. Without any loss of generality, we may write this relationship as

$$f_1(Q_1, Q_2, \dots, Q_n) = 0 \quad (1)$$

Since physical quantities are measured in physical units (feet, seconds, lbs, degrees, etc.) or some algebraic combination of units, it can be shown that the above general relationship must have the form

$$\sum M Q_1^{b_1} Q_2^{b_2} \dots Q_n^{b_n} = 0$$

where M is a dimensionless number and each of the products being summed must have the same units (otherwise you could not add them together). If we divide both sides by any one term in the sum, the expression takes the form

$$\sum N Q_1^{a_1} Q_2^{a_2} \dots Q_n^{a_n} + 1 = 0$$

where each product in the sum is a dimensionless quantity. If we represent each of these terms by π_i , then we may rewrite this equation as

$$f_2(\pi_1, \pi_2, \dots, \pi_n) = 0 \quad (2)$$

Loosely stated, the Buckingham π -Theorem declares that the number of dimensionless products π_i necessary for Equation (2) to represent Equation (1) is $n - k$, where k is the number of fundamental dimensions used to define the variables.

Applying the above principles, let us assume that we have a single spool turbojet engine having the following independent parameters (and dimensions)

Parameter	Dimension
N spool speed (rpm) 1/t	
T_1 ambient temperature	T
P_1 ambient pressure	M/(Lt ²)
v aircraft velocity	L/t
D compressor diameter	L
R gas constant	L ² /(t ² T)

and dependent variable

$$Fn \quad \text{Thrust} \quad ML/t^2$$

where t =Time, T =temperature (degrees), M =Mass, and L =Length. The total number of fundamental dimensions in this case is 4.

Thus if we assume that we have a relationship of the form $Fn = f(N, T_2, P_2, v, D, R)$, which implies an implicit form $f(Fn, N, T_2, P_2, v, D, R) = 0$, involving 7 quantities. By virtue of the π -theorem, we may express the relationship in terms of $7-4=3$ dimensionless quantities as follows:

$$\phi(\pi_1, \pi_2, \pi_3) = 0$$

where

$$\pi_1 = N P_2^b T_2^c D^d R^e, \pi_2 = v P_2^e T_2^f D^g R^h, \pi_3 = F_n P_2^k T_2^m D^n R^r$$

The π_i terms are dimensionless and hence their units must be equal to $L^0 M^0 T^0$. For example

$$\begin{aligned} \pi_1 &= \left(\frac{1}{t}\right) \left(\frac{M}{L t^2}\right)^b T^c \left(\frac{L^2}{t^2 T}\right)^d = L^0 M^0 T^0 \\ &= L^{c+2d-a} M^a T^{b-d} t^{-1-2a-2d} \end{aligned}$$

$\therefore$

$$0 = a$$

$$0 = c + 2d - a$$

$$0 = b - d$$

$$0 = 1 + 2a + 2d \quad \text{which implies} \quad a = 0$$

$$b = d = -1/2$$

$$c = -1$$

This yields a parameter of the form

$$N / (L \sqrt{RT_2})$$

Repeating this process for the other dimensionless quantities, yields the following relationship

$$\frac{F}{P_2 A} = f_1 \left(\frac{N}{D \sqrt{RT_2}}, \frac{v}{\sqrt{RT_2}} \right)$$

and if we assume a constant geometry ($A=\text{const}$, $D=\text{const}$) we obtain

$$\frac{F}{P_1} = f_2 \left(\frac{N}{\sqrt{T_2}}, Mn \right) \Rightarrow \frac{F}{\delta} = f_3 \left(\frac{N}{\sqrt{\theta}}, Mn \right)$$

which would imply that the appropriate corrections for net thrust and spool speed are F/δ and $N/\sqrt{\theta}$ respectively.

INTRODUCTION

In the sequel, we will assume a two shaft, mixed flow, turbofan engine with station numbers as depicted in Figure 1 below. This reference is made for convenience only, and all corrections that are given in this paper can be applied equally to all gas turbine engines.

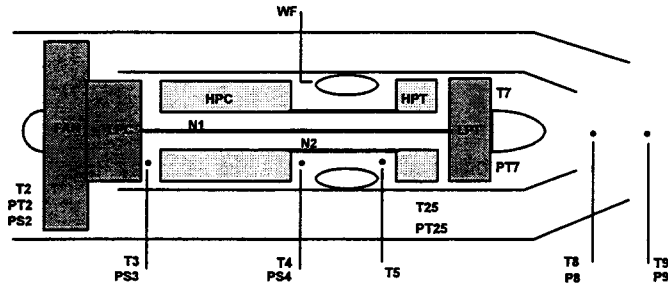


FIGURE 1: TWIN SPOOL MIXED FLOW TURBOFAN

For any gas path parameter $\mathcal{P}$, the equivalent corrected parameter will be denoted by $\mathcal{P}^*$ throughout this discussion. In general, a change in the engine inlet conditions T_2 and P_2 will be accompanied by an attendant change in any downstream gas path parameter $\mathcal{P}$. A corrected parameter $\mathcal{P}^*$ would be constant regardless of the change in inlet condition and represents the value the parameter $\mathcal{P}$ would have at some fixed reference inlet condition. This reference condition can be whatever you like, however, it is common practice to select *standard day* conditions ($T_2 = 518.67$ deg R, $P_2 = 14.696$ psia) for this purpose. Thus, we can assume without loss of generality that $\mathcal{P} = f(T_2, P_2, \mathcal{P}^*)$. Thus it follows that

$$d\mathcal{P} = \left(\frac{\partial \mathcal{P}}{\partial T_2}\right) dT_2 + \left(\frac{\partial \mathcal{P}}{\partial P_2}\right) dP_2 + \left(\frac{\partial \mathcal{P}}{\partial \mathcal{P}^*}\right) d\mathcal{P}^*$$

$$\frac{d\mathcal{P}}{\mathcal{P}} = \left(\frac{\partial \mathcal{P} / \mathcal{P}}{\partial T_2 / T_2}\right)_{P_2=\text{const}, \mathcal{P}^*=\text{const}} \frac{dT_2}{T_2} + \left(\frac{\partial \mathcal{P} / \mathcal{P}}{\partial P_2 / P_2}\right)_{T_2=\text{const}, \mathcal{P}^*=\text{const}} \frac{dP_2}{P_2} + \left(\frac{\partial \mathcal{P} / \mathcal{P}}{\partial \mathcal{P}^* / \mathcal{P}^*}\right)_{T_2=\text{const}, P_2=\text{const}} \frac{d\mathcal{P}^*}{\mathcal{P}^*}$$

Now if we make the simplifying assumption that the first two partials are constant (the third clearly is constant and equal to unity) and denote them by a and b respectively, then

$$\frac{d\mathcal{P}}{\mathcal{P}} \approx a \frac{dT_2}{T_2} + b \frac{dP_2}{P_2} + \frac{d\mathcal{P}^*}{\mathcal{P}^*}$$

If we now define, the dimensionless parameters $\theta = T_2/518.67$ and $\delta = P_2/14.696$, then we arrive at

$$\frac{d\mathcal{P}^*}{\mathcal{P}^*} \approx \frac{d\mathcal{P}}{\mathcal{P}} - a \frac{d\theta}{\theta} - b \frac{d\delta}{\delta} \Rightarrow \mathcal{P}^* \approx \frac{\mathcal{P}}{\theta^a \delta^b} \quad (3)$$

Thus, a corrected parameter maintains the same units and is calculated by means of theta and delta exponent corrections. We might mention at this juncture, that Equation (3) is not only approximate for the reasons already alluded to, but in addition, some gas path parameters

such as fuel flow (W_f), for example, are also corrected for humidity and fuel heating value. The impact of changes in viscosity with altitude (Reynolds effects) or changes in the gas composition and their impact on gas path parameters will not be considered in this paper. A discussion of these types of corrections are beyond the scope of this presentation and thus, we will assume the above form (Eq. 3) for all corrected parameters $\mathcal{P}^*$.

The values for a and b will, in general, vary with engine type and cycle, however, there are some values which might be considered standard classical corrections and are approximations which are commonly used in practice for all gas turbines. Table 1 depicts many of the common gas path parameters and their standard day corrections. In the sequel we will attempt to develop all of these and show the classical value as a special case, where applicable.

Table 1: Common Gas Turbine Parameter Corrections

Parameter	a	b	Corrected Symbol
Rotor Speed	0.5	0	$N^* = \frac{N}{\sqrt{\theta}}$
Airflow	-0.5	1	$W_a^* = \frac{W_a \sqrt{\theta}}{\delta}$
Fuelflow	a (classical: 0.5)	1	$W_f^* = \frac{W_f}{\theta^a \delta} = \frac{W_f}{\sqrt{\theta} \delta}$
Thrust	0	1	$F_n^* = \frac{F_n}{\delta}$
HorsePower	0.5	1	$HP^* = \frac{HP}{\sqrt{\theta} \delta}$
Torque	0	1	$Q^* = \frac{Q}{\delta}$
Temperature	a (classical: 1.0)	0	$T^* = \frac{T}{\theta^a} = \frac{T}{\theta}$
Pressure	0	1	$P^* = \frac{P}{\delta}$
Acceleration	0	1	$\dot{N}^* = \frac{\dot{N}}{\delta}$
Metal Temp Rate	0.74	0.8	$\dot{T}_m^* = \frac{\dot{T}_m}{\theta^{0.74} \delta^{0.8}}$

DERIVATIONS

In the sequel, we shall attempt to supply some rationale for these corrections. Most will be motivated from simple thermodynamic relationships and simplifying assumptions and do not require an extensive knowledge of either thermodynamics or gas turbine operation. Where applicable we will attempt to supply the required definitions.

The most extensively used and perhaps the simplest correction of all is the correction for temperatures, and is the first one which we shall consider.

Corrected Temperature: T/θ

As a general rule, it seems reasonable that engine inlet temperature (T_2) should have a direct effect on downstream temperatures in the engine's gas path. It seems natural, for example, that at a given operating condition, increasing T_2 would in turn increase T_3 , which in turn would increase T_4 , etc. To motivate this concept, we need to consider the T-S diagram. To be definite, let us consider the compression induced by the Fan and LPC from station 2 to station 3 of our sample engine in Figure 1. Without loss of generality, an isentropic compression has been assumed. Figure 2 depicts a compression from P_2 to P_3 starting at different temperatures (T_2 and T_2^*).

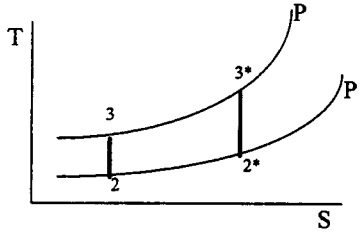


Figure 2: Compression T-S Diagram

At this point we will note two properties of the T-S diagram which we will state without proof. (The reader can derive these for him/herself, armed only with the elementary laws of thermodynamics.) The relationships are

1. The curves of constant pressure are monotonically increasing with S
2. The constant pressure curves diverge from one another with increasing S

These properties suggest that the * quantities are larger in magnitude than their non-starred counterparts and that $(T_3 - T_2) < (T_3^* - T_2^*)$. What we will now show, however, is that the ratio of T_3 over T_2 is approximately equal to T_3^* over T_2^* !

To accomplish this we need one more relationship from basic thermodynamics (which follows from the definition of entropy and the perfect gas law).

$$dS = \frac{dh}{T} - R \frac{dP}{P}$$

where S, h, T, P, R denote entropy, enthalpy, temperature, pressure and the gas constant, respectively. Integrating this relationship between states 2 and 3 and assuming constant specific heat, yields,

$$\begin{aligned} 0 &= \int_2^3 dS = \int_2^3 \frac{dh}{T} - R \int_2^3 \frac{dP}{P} = c_p \int_2^3 \frac{dT}{T} - R \int_2^3 \frac{dP}{P} \\ &= c_p \ln\left(\frac{T_3}{T_2}\right) - R \ln\left(\frac{P_3}{P_2}\right) \end{aligned}$$

$\Rightarrow$

$$\frac{T_3}{T_2} = \left(\frac{P_3}{P_2}\right)^{R/c_p}$$

Likewise, integrating between 2* and 3* with the same assumption on specific heats, provides

$$\begin{aligned} \frac{T_3^*}{T_2^*} &= \left(\frac{P_3}{P_2}\right)^{R/c_p} \\ &= \frac{T_3}{T_2} \end{aligned}$$

$\Rightarrow$

$$T_3^* = \frac{T_3}{(T_2/T_2^*)} = \frac{T_3}{\theta}$$

as required.

Corrected Pressure: P/δ

The pressure changes experienced at various stations throughout the engine's gas-path are the effect of either compressions or expansions resulting from the action of the engine's turbomachinery. For purposes of motivating the correction for pressure, we can consider without loss of generality, a compression process; the argument for expansion pressures would be similar.

Let us consider, for example, the pressure P_3 at the exit of the LPC. This pressure is related to temperature and pressure by virtue of the following relationship

$$\frac{T_3}{T_2} = \left(\frac{P_3}{P_2}\right)^{\frac{k-1}{k\eta}} \quad \text{where } \eta = \text{polytropic efficiency}$$

$$k = c_p / c_v = \text{ratio of specific heats}$$

Thus, since $T_3/\theta = \text{constant}$ implies $T_3/T_2 = \text{constant}$, we have that $P_3/P_2 = \text{constant}$, which in turn implies that $P_3/\delta = \text{constant}$, which establishes the correction at station 3. Implicit in this argument is that the ratio of specific heats k and efficiency η do not change. Moving downstream to station 4 we have that

$$\frac{P_4}{P_3} = \left(\frac{T_4}{T_3} \right)^{\frac{k\eta}{k-1}} = \left(\frac{T_4/\theta}{T_3/\theta} \right)^{\frac{k\eta}{k-1}} = \text{const.}$$

∴

$$\frac{P_4}{\delta} = \frac{P_4}{P_3} \frac{P_3}{\delta} = \text{const}$$

Thus, we proceed downstream and establish in general that $P/\delta = \text{constant} = P^*$.

Corrected Mass Flow: $W_a \sqrt{\theta} / \delta$

Consider flow W_a (in pps) moving with uniform velocity v through a pipe having cross sectional area A (ft²). Then

$$W_a = \rho A v = \text{density (lbs / ft}^3) \times \text{area (ft}^2) \times \text{velocity (ft / sec)}$$

For an ideal gas, we also know that $P_s = \rho RT$, from which it follows that

$$\frac{W_a}{A} = \rho v = \frac{P_s}{RT_s} v$$

and from the definition of mach number M_n

$$M_n = v / \sqrt{gkRT_s}$$

we obtain the relationship

$$\frac{W_a}{A} = \frac{P_s M_n \sqrt{gkRT_s}}{RT_s} \Rightarrow \frac{W_a \sqrt{T_t}}{AP_t} = \left(\frac{P_s}{P_t} \right) \sqrt{\frac{T_t}{T_s}} M_n \sqrt{\frac{gk}{R}}$$

We know, however, that for an isentropic process

$$\frac{T_t}{T_s} = 1 + \frac{k-1}{2} M_n^2 \quad \text{and} \quad \frac{P_t}{P_s} = \left(1 + \frac{k-1}{2} M_n^2 \right)^{\frac{k}{k-1}}$$

Substituting, we obtain

$$\frac{W_a \sqrt{T_t}}{AP_t} = \sqrt{\frac{gk}{R}} M_n \left(1 + \frac{k-1}{2} M_n^2 \right)^{\frac{k+1}{2(1-k)}}$$

and multiplying both sides by $(\theta/T_t)^{1/2} / (\delta/P_t)$ to get the proper units (pps) yields

$$\frac{W_a \sqrt{\theta}}{\delta} = \frac{A}{P_t / \delta} \sqrt{\frac{gk(T_t/\theta)}{R}} M_n \left(1 + \frac{k-1}{2} M_n^2 \right)^{\frac{k+1}{2(1-k)}} = \text{constant}$$

for $M_n = \text{const.}$, thus providing a *corrected* quantity.[†]

Corrected Fuel Flow: $W_F / \delta \sqrt{\theta}$

Consider the (simplified) energy equation bounding the combustor

$$\frac{BTU}{\text{sec}} = \eta_b q W_F = (W_a + W_F) \Delta h = (W_a + W_F) (h_5 - h_4)$$

where h denotes enthalpy, q denotes the heating value of the fuel and η denotes the adiabatic efficiency of the combustor. If we assume that (at a given operating point) efficiency and fuel heating value are constant, then if we take logs of both sides and differentiate, we obtain

$$\begin{aligned} \frac{dW_F}{W_F} &= \frac{d(W_a + W_F)}{(W_a + W_F)} + \frac{d(\Delta h)}{\Delta h} \\ &= \left(\frac{W_a}{(W_a + W_F)} \right) \frac{dW_a}{W_a} + \left(\frac{W_F}{(W_a + W_F)} \right) \frac{dW_F}{W_F} \\ &\quad + \left(\frac{c_{p5} T_5}{\Delta h} \right) \frac{dT_5}{T_5} - \left(\frac{c_{p4} T_4}{\Delta h} \right) \frac{dT_4}{T_4} \end{aligned}$$

and a little algebraic manipulation gives

$$\left(\frac{W_a}{W_a + W_F} \right) \frac{dW_F}{W_F} = \left(\frac{W_a}{(W_a + W_F)} \right) \frac{dW_a}{W_a} + \left(\frac{c_{p5} T_5}{\Delta h} \right) \frac{dT_5}{T_5} - \left(\frac{c_{p4} T_4}{\Delta h} \right) \frac{dT_4}{T_4}$$

Now, at a given operating point, we can assume that the following relationships hold:

$$\begin{aligned} \frac{W_a \sqrt{\theta}}{\delta} &= \text{const} \Rightarrow \frac{dW_a}{W_a} + 1/2 \frac{dT_4}{T_4} - \frac{dP_4}{P_4} = 0 \\ \frac{T_4}{\theta} &\text{ and } \frac{T_5}{\theta} = \text{const} \Rightarrow \frac{dT_4}{T_4} = \frac{dT_2}{T_2} = \frac{dT_5}{T_5} \\ \frac{P_4}{\delta} &= \text{const} \Rightarrow \frac{dP_4}{P_4} = \frac{dP_2}{P_2} \end{aligned}$$

Thus,

$$\begin{aligned} \frac{dW_F}{W_F} &= \frac{dP_2}{P_2} - 1/2 \frac{dT_2}{T_2} + \left(\frac{W_a + W_F}{W_a} \frac{c_{p5} T_5}{\Delta h} \right) \frac{dT_2}{T_2} - \left(\frac{W_a + W_F}{W_a} \frac{c_{p4} T_4}{\Delta h} \right) \frac{dT_2}{T_2} \\ &= \frac{dP_2}{P_2} + \underbrace{\left[\left(1 + \frac{W_F}{W_a} \right) \left(\frac{c_{p5} T_5}{\Delta h} - \frac{c_{p4} T_4}{\Delta h} \right) - 1/2 \right]}_x \frac{dT_2}{T_2} \quad (5) \\ &= \frac{dP_2}{P_2} + x \frac{dT_2}{T_2} \\ &\Rightarrow \frac{W_F}{\delta \theta^x} = \text{const} \end{aligned}$$

[†] T/θ and P/δ are corrected parameters and hence constant by virtue of the previous derivations.

Furthermore, if we make the assumption that specific heats are constant, i.e. $c_{p4} = c_{p5}$ then the exponent x reduces to $1/2 + \text{Fuel-Air Ratio} = (\text{approximately}) 1/2$ and we obtain the classical normalization

$$W_F^* \approx \frac{W_F}{\delta \sqrt{\theta}}$$

Corrected Net Thrust: F_n / δ

Expressions for the net thrust produced by a gas turbine engine will depend upon the configuration of the engine, i.e., whether it is a turbojet or a turbofan, mixed flow or non-mixed flow, as well as an expression for ram drag as a function of mach number. To keep the derivation as simple as possible, we will assume a static thrust expression, i.e., mach = 0, no ram drag and, we will once again refer to our sample engine configuration which is that of a mixed flow turbofan. It should be noted that derivation that follows can be repeated (in spirit) with more complicated expressions for net thrust for other engine types and flight conditions to arrive at similar representations for corrected thrust.

The thrust produced by this engine depends on the momentum of the air ejected from the exhaust nozzle as well as the net force (due to pressure) acting across the total area of the nozzle. In symbols

$$F_n = \frac{W_9}{g} v_9 + A_9 (P_9 - P_{s9})$$

where

$W_9 = W_8 =$.Total airflow at nozzle

$A_9 =$.Nozzle Area

$P_9 =$.Nozzle exit total pressure

$P_{s9} = P_2 =$.Nozzle exit static pressure (ambient)

$v_9 = \sqrt{2gJ(h_9 - h_{s9})} =$.Velocity at exhaust nozzle

$h_9 =$.Total (stagnation) enthalpy at exit

$h_{s9} =$.Static enthalpy at exit

$J =$.Mechanical equivalent of heat

.(778 ft - lbs / BTU)

The expression for the exit velocity follows directly from the definition of stagnation (total) enthalpy. Similarly, there is an expression for the velocity at station 8 (mixer) involving total and static enthalpies at station 8. The flow from station 8 to 9 (through a nozzle) we will assume to be adiabatic, i.e., no work done, no heat added. By conservation of energy, the total enthalpies are equal ($h_8 = h_9$). If we denote the change in enthalpy ($h_9 - h_{s9}$) = ($h_8 - h_{s9}$) = Δh , then we may rewrite the above expression for thrust as

$$F_n = \frac{W_8}{g} \sqrt{2gJ\Delta h} + A_9 (P_9 - P_{s9})$$

Taking logs of both sides and differentiating, produces

$$\begin{aligned} \frac{dF_n}{F_n} &= \frac{(W_8/g) \sqrt{2gJ\Delta h}}{F_n} \left[\frac{dW_8}{W_8} + \frac{1}{2} \frac{d\Delta h}{\Delta h} \right] \\ &+ \frac{A_9 (P_9 - P_{s9})}{F_n} \left[\frac{dA_9}{A_9} + \left(\frac{P_9}{P_9 - P_2} \right) \frac{dP_9}{P_9} - \left(\frac{P_2}{P_9 - P_2} \right) \frac{dP_2}{P_2} \right] \\ &= \frac{(W_8/g) \sqrt{2gJ\Delta h}}{F_n} \left[\frac{dW_8}{W_8} + \left(\frac{h_8}{2\Delta h} \right) \frac{dh_8}{h_8} - \left(\frac{h_{s9}}{2\Delta h} \right) \frac{dh_{s9}}{h_{s9}} \right] \\ &+ \frac{A_9 (P_9 - P_{s9})}{F_n} \left[\underbrace{\frac{dA_9}{A_9}}_{\text{zero (fixed geometry)}} + \left(\frac{P_9}{P_9 - P_2} \right) \frac{dP_9}{P_9} - \left(\frac{P_2}{P_9 - P_2} \right) \frac{dP_2}{P_2} \right] \\ &= \frac{(W_8/g) \sqrt{2gJ\Delta h}}{F_n} \left[\frac{dW_8}{W_8} + \left(\frac{c_{p8} T_8}{2\Delta h} \right) \frac{dT_8}{T_8} - \left(\frac{c_{p9} T_{s9}}{2\Delta h} \right) \frac{dT_{s9}}{T_{s9}} \right] \\ &+ \frac{A_9 (P_9 - P_{s9})}{F_n} \left[\left(\frac{P_9}{P_9 - P_2} \right) \frac{dP_9}{P_9} - \left(\frac{P_2}{P_9 - P_2} \right) \frac{dP_2}{P_2} \right] \end{aligned}$$

Now, at a given engine operating point (steady state), the following conditions hold

$$\begin{aligned} \frac{W_8 \sqrt{\theta_8}}{\delta_8} &= \text{constant} \Rightarrow \frac{dW_8}{W_8} + \frac{1}{2} \frac{dT_8}{T_8} - \frac{dP_8}{P_8} = 0 \\ \frac{T_8}{\theta_2} &= \text{constant} \Rightarrow \frac{dT_8}{T_8} - \frac{dT_2}{T_2} = 0 \\ \frac{T_{s9}}{\theta_2} &= \text{constant} \Rightarrow \frac{dT_{s9}}{T_{s9}} - \frac{dT_2}{T_2} = 0 \\ \frac{P_8}{\delta_2} &= \text{constant} \Rightarrow \frac{dP_8}{P_8} - \frac{dP_2}{P_2} = 0 \\ \frac{P_9}{\delta_2} &= \text{constant} \Rightarrow \frac{dP_9}{P_9} - \frac{dP_2}{P_2} = 0 \end{aligned}$$

Substituting, we obtain

$$\begin{aligned} \frac{dF_n}{F_n} &= \frac{(W_8/g) \sqrt{2gJ\Delta h}}{F_n} \left[\frac{dP_8}{P_8} + \underbrace{\left(\frac{c_{p8} T_8}{2\Delta h} - \frac{c_{p9} T_{s9}}{2\Delta h} - \frac{1}{2} \right)}_x \frac{dT_2}{T_2} \right] \\ &+ \frac{A_9 (P_9 - P_{s9})}{F_n} \frac{dP_2}{P_2} \\ &= \underbrace{\frac{(W_8/g) \sqrt{2gJ\Delta h}}{F_n}}_{y_1} \left[\frac{dP_8}{P_8} + x \frac{dT_2}{T_2} \right] + \underbrace{\frac{A_9 (P_9 - P_{s9})}{F_n}}_{y_2} \frac{dP_2}{P_2} \end{aligned}$$

$$= (y_1 + y_2) \frac{dP_2}{P_2} + y_1 x \frac{dT_2}{T_2} = \frac{dP_2}{P_2} + y_1 x \frac{dT_2}{T_2}$$

$$\Rightarrow \frac{F_n}{\delta_2 \theta_2^{y_1 x}} = \text{constant}$$

Since the specific heats at station 8 and 9 are approximately equal (since $T_8 \approx T_{9s}$), we may assume a common specific heat c_p . In this case, the theta exponent x reduces to 0 (since $\Delta h = c_p \Delta T$). Thus, we arrive at the classical correction for thrust, namely

$$F_n^* = \frac{F_n}{\delta_2}$$

Corrected Horsepower: $HP / \delta \sqrt{\theta}$

Horsepower required (or developed) by a compressor (or turbine) is given by

$$HP = \frac{J}{550} W_a c_p \Delta h \approx \frac{J}{550} W_a c_p \Delta T \text{ where } J = 778.17 \text{ ft lb / BTU}$$

A *corrected* horsepower will not vary with inlet temperature and pressure and thus can be written as above with corrected flow and temperature, i.e.

$$HP^* = \frac{J}{550} W_a^* c_p \Delta h^* \approx \frac{J}{550} W_a^* c_p \Delta T^*$$

$$= \frac{J}{550} \frac{W_a \sqrt{\theta}}{\delta} c_p \frac{\Delta T}{\theta}$$

$$= \frac{J}{550} \frac{W_a c_p \Delta T}{\delta \sqrt{\theta}}$$

$$= \frac{HP}{\delta \sqrt{\theta}}$$

Corrected Rotational Speeds: $N / \sqrt{\theta}$

Tangential velocity is related to rotational speed (rpm) by the radius of the object (compressor or turbine blade) in question. It is also related to acoustic velocity by Mach number and the square root of temperature.

Mathematically, we have

$$v = r N = M_n \sqrt{g k R T}$$

where v is the velocity in ft/s, r is the radius (ft), N is the rotational speed (rpm), g , k , and R are constants (gravity, ratio of specific heats and the gas constant respectively), and M_n is Mach number. Taking logs and differentiating, we obtain

$$\frac{dN}{N} = \frac{dM_n}{M_n} + \frac{1}{2} \frac{dT}{T} = \frac{1}{2} \frac{dT}{T} \text{ since Mach} = \text{constant}$$

$$\Rightarrow \frac{dN}{N} - \frac{1}{2} \frac{dT}{T} = 0$$

$$\Rightarrow \frac{N}{\sqrt{\theta}} = \text{constant}$$

Corrected Torque: Q / δ

The correction for torque follows from corrections for speed and horsepower, since horsepower is the product of torque and speed. It follows that a corrected horsepower should be a product of corrected torque and corrected speed, i.e.

$$HP^* = Q^* N^*$$

$$= \frac{HP}{\delta \sqrt{\theta}} = Q^* \frac{N}{\sqrt{\theta}}$$

$$\Rightarrow Q^* = \frac{HP}{N \delta} = \frac{Q}{\delta}$$

Corrected Acceleration $\dot{N}$ dot: $\dot{N} / \delta$

From Newton's Law, Torque (force) = Moment of Inertia (I) $\times$ Acceleration:

$$Q = I \dot{N}$$

$$\therefore Q^* = I \dot{N}^* \Rightarrow \frac{Q}{\delta} = I \frac{\dot{N}}{\delta}$$

$$\Rightarrow \dot{N}^* = \dot{N} / \delta$$

Corrected Metal Temperature Rate $\dot{T}_m$ dot: $\frac{\dot{T}_m}{\theta^{.74} \delta^{.8}}$

The (time) rate of change of a metal temperature is typically modeled as a 1st order lag heat transfer between the gas path temperature T and the metal temperature T_m . In symbols,

$$\dot{T}_m = \frac{1}{\tau} (T - T_m) \quad (3)$$

where time constant $\tau = \frac{m c_p}{H A}$

The McAdams correlation for turbulent flow over a flat plate (Shepherd, p254) is given by

$$Nu = 0.023(Re)^{0.8} (Pr)^{0.4}$$

where Nu, Re, and Pr are the Nusselt's, Reynold's and Prandtl's numbers respectively which are defined as follows:

$$\text{Nu} = \frac{HD}{\kappa}, \quad \text{Re} = \frac{\rho v D}{\mu}, \quad \text{Pr} = \frac{c_p \mu}{\kappa}$$

Substituting and re-arranging terms we can obtain the following form:

$$\begin{aligned} H &= 0.023 \frac{\kappa}{D} \left(\frac{\rho v D}{\mu} \right)^{0.8} \left(\frac{c_p \mu}{\kappa} \right)^{0.4} \\ &= 0.023 \frac{\kappa^{0.6} c_p^{0.4}}{D^{0.2}} \frac{\rho^{0.8} v^{0.8}}{\mu^{0.4}} \\ &= \left(\frac{0.023 \kappa^{0.6}}{c_p^{0.6} \mu^{0.6}} \right) \frac{c_p \mu^{0.2} \rho^{0.8} v^{0.8}}{D^{0.2}} \\ &= \left(\frac{0.023}{\text{Pr}^{0.6}} \right) \frac{c_p \mu^{0.2} \rho^{0.8} v^{0.8}}{D^{0.2}} \\ &\approx 0.027 \frac{c_p \mu^{0.2} \rho^{0.8} v^{0.8}}{D^{0.2}} \end{aligned} \quad (4)$$

where the final approximation is obtained by noting that Prandtl's number is approximately constant in the range of interest (Shepherd, p 254). At this juncture we may recall that mass flow $W_a = \rho A v$ was corrected by square root of theta over delta. Combining these observations, we have

$$\frac{W_a}{A} = \frac{W_a^* \delta}{\sqrt{\theta}} = \rho v$$

Therefore,

$$(\rho v)^{0.8} = (W_a^*)^{0.8} \delta^{0.8} \theta^{-0.4}$$

Using the approximation at the end of equations (4) to define the reference condition H^* and noting that $D=D^*$, we can write the ratio

$$\begin{aligned} \frac{H^*}{H} &= \left(\frac{c_p^*}{c_p} \right) \left(\frac{\mu^*}{\mu} \right)^{0.2} \left(\frac{\rho^* v^*}{\rho v} \right)^{0.8} \\ &= \left(\frac{c_p^*}{c_p} \right) \left(\frac{1}{\theta^n} \right)^{0.2} \left(\frac{\sqrt{\theta}}{\delta} \right)^{0.8} \\ &= \left(\frac{c_p^*}{c_p} \right) \theta^{0.4-0.2n} \delta^{-0.8} \end{aligned}$$

where we have assumed that the ratio μ^*/μ is solely a function of temperature (θ). Now using equation (3) to define $\dot{T}_m^*$ in a similar manner, we can form the appropriate ratio and use the above relationship to simplify the expression.

$$\begin{aligned} \frac{\dot{T}_m^*}{\dot{T}_m} &= \left(\frac{H^*}{H} \right) \left(\frac{c_p}{c_p^*} \right) \frac{T^* - T_m^*}{T - T_m} \\ &= \theta^{0.4-0.2n} \delta^{-0.8} \frac{(T/\theta) - (T_m/\theta)}{T - T_m} \\ &= \theta^{-0.6-0.2n} \delta^{-0.8} \\ &\Rightarrow \\ \dot{T}_m^* &= \frac{\dot{T}_m}{\theta^{0.6+0.2n} \delta^{0.8}} \end{aligned}$$

The value of n appearing in the theta exponent can be determined experimentally by correlating viscosity with temperature from which it is observed that n varies from 0.8 at low temperature to 0.6 at high temperature. These values provide a theta exponent ranging from 0.76 to 0.72, respectively. The average value of 0.74 is recommended, thus providing the requisite correction.

$$\dot{T}_m^* = \frac{\dot{T}_m}{\theta^{0.74} \delta^{0.8}}$$

DETERMINING CORRECTION FACTORS

The methods for obtaining appropriate corrections for gas path parameters depends in part on the application at hand as well as the amount and type of information (data) available. These methods can be split into essentially two groups: 1) *analytical* and 2) *empirical*. In practice it is more than likely that a combination of both types of techniques will be utilized. We will briefly illustrate both of these approaches.

Analytical Methods

As a practical application of how to utilize some of the above information to determine a theta correction for Wf in a turbo-fan engine from engine simulation information, consider the information contained in Table 2. This data was obtained from a non-linear aerothermodynamic representation of a small turbofan engine. In order to avoid introducing new station numbers, we will use the corresponding station numbers of our sample engine in Figure 1. The thermodynamic deck provides the following information at station 4 (HPC exit) and station 5 (HPT inlet) at one engine operating point.

Applying these values to the Wf Equation (5) yields a value of 0.707 for the theta exponent " x ". The actual value used for this engine is 0.69. Presumably, the above calculation would yield slightly different results at differing engine operating points, the average of which is probably around 0.69.

Table 2: Non-linear Engine Simulation Data

Parameter		Station 4	Station 5	Units
k		1.39801	1.30214	ft-lb/deg R
h		300.297	657.18	BTU
T		1235.18	2436.52	deg R
W_a		23.2613	23.7252	pps
W_f	0.4639			pps
Δh	356.883			
W_f / W_a	0.019943			
cp		0.24081	0.29542	
$cp T / \Delta h$		0.83346	2.01690	
$1 + W_f / W_a$	1.019943			
Exponent	0.707			

This type of analytic approach could be made by applying some of the intermediate equations used in the derivations for various gas path parameters in much the same way as in the above example. Another analytic method would be to construct a complete differential model of the engine [Volponi and Urban, 1992]. Models of this type provide sensitivities to % changes in independent parameters (T_2 is one such) to corresponding % changes in dependent engine parameters (such as W_f , for example). The difference between this and what was done in the derivation of the W_f correction, is that the latter would be one of many differential equations (usually around 40) that would relate *all* the independent and dependent parameters at various engine stations interrelated through the laws of thermodynamics, energy and flow conservation. The simultaneous solution of these 40 equations would produce (for example) a sensitivity of T_2 on spool speeds (N), fuel flow (W_f), temperatures at various stations, etc. These sensitivities give an indication of the (theta) correction to be applied.

An example of such an approach is illustrated in the sensitivity matrix in Table 3 below.

The sensitivities of engine inlet temperature and pressure are given in the last two columns for all of the dependent parameters listed in the 1st column. For example, the dependent parameter N1C2 which is *corrected* low spool speed has a sensitivity of -0.0105 with Tam. This means that the following is true:

$$\frac{dN1c2}{N1c2} = \frac{dN1}{N1} - \frac{1}{2} \frac{dT2}{T2} = -0.0105 \frac{dT2}{T2} \Rightarrow \frac{dN1}{N1} - 0.4895 \frac{dT2}{T2} = 0$$

which implies that the correction for N1 is $N1/0.4895$ instead of $N1/0.5$. Likewise, the correction for N2 is .4833 instead of square root. Now if we choose some raw parameters such as T3, T4, and T7, we see that the suggested theta exponents are 0.9816, 0.9291, and 0.9746, respectively, and not 1.0 as the classical correction would dictate. Fuel flow is demanding an exponent of 0.6014, instead of

0.5. All of these theta exponents derivable from a differential model will provide a correction that is closer to the truth than the classical (providing the model is reasonably correct) and can be used as a good *first guess* for the correction to be used. The fidelity of these corrections will of course depend on the fidelity of the underlying differential model. For general analysis, classical corrections often suffice. Several parameters may require a finer correction. W_f is typically one of those parameters and it is not uncommon to see theta exponents ranging from 0.5 to 0.7 for this parameter. Such a refinement is required for work demanding greater accuracies, such as diagnostics, and for these purposes it is customary to use corrections such as those given in the sensitivity matrix above, and when actual engine data becomes available, to adjust these first guesses empirically. This leads us to our next topic.

Table 3: Sensitivity Matrix

	TAM	PAM
N1C2	-0.0105	0.0077
N2C2	-0.0167	0.0021
WF	0.6014	0.9933
FN	0.0123	1.0035
P3/P2	-0.0119	0.007
P4/P3	-0.014	-0.003
P7/P2	0	0
T3	0.9816	0.0037
T4	0.9291	0.0025
T5	0.9265	-0.005
T6	0.9542	-0.005
T7	0.9746	-0.006
P3	-0.0119	1.007
P4	-0.0259	1.0037
P5	-0.0224	1.0031
P6	-0.0073	1.0024
WA	-0.5076	1.0056
T2	1	0
P2	0	1

Empirical Methods

Empirical methods utilize actual engine data collected over a temperature range and comprise more of a statistical approach to the subject of parameter normalization and as such, do not require any in-depth knowledge of gas turbines or their operation. It has become a generally accepted practice among engine performance diagnosticians to perform some sort of empirical analysis to verify the validity of the correction factors being used. Neglecting to do so exposes the diagnostic procedure to greater error which may render the diagnostics entirely useless.

There are many techniques that can be put forward to analyze engine data for normalization purposes, however, we will restrict our discussion to a simple formula (derived by this author, circa 1980), suggest several alternative approaches, and finally offer some Monte Carlo simulation results that suggest that all of the above produce roughly the same result.

For simplicity, we will consider only temperature (θ) corrections; pressure (δ) corrections can be handled in an analogous manner. Let $P^* = P/\theta^x$, where x denotes the true (sought) exponent and let α be a theta exponent used to normalize a set of data $\{P_i\}_{i=1,\dots,n}$. Let us further assume that each datum P_i (for all i) was obtained at an inlet condition represented by θ_i and that we have available, the nominal value of this parameter for the engine operating condition(s) at which the parameters P_i were obtained. Let us denote this *uncorrected* nominal value by P_{NOM} . Then

$$\frac{P_i}{\theta^\alpha} = \frac{P_{NOM}}{\theta^x} \left(1 + \frac{\Delta G_i}{100}\right)$$

where ΔG_i represents the % of point delta from nominal (for all i). Now if we plot these deltas against θ , then if α were the true correction exponent, we would expect to see a scatter plot that would have zero correlation, i.e. a linear regression line fit horizontal with the theta axis. An upward or downward trend, however, would indicate the extent to which the exponent α was deficient. If we let m and b , denote the regression slope and intercept respectively, then we have a defined linear relationship as follows:

$$\begin{aligned} \Delta G &= m \theta + b \quad \text{where} \quad b = \Delta \bar{G} - m \bar{\theta} \\ &= \frac{1}{n} \sum_{i=1}^n \Delta G_i - m \left[\frac{1}{n} \sum_{i=1}^n \theta_i \right] \\ \text{and} \quad m &= r \frac{s_{\Delta G}}{s_\theta} \end{aligned}$$

where r is the correlation coefficient and $s_{\Delta G}$ and s_θ are the sample standard deviations for ΔG and θ , respectively. Thus for arbitrary θ , we have

$$\begin{aligned} \frac{P}{P_{NOM}} \theta^{x-\alpha} &\approx 1 + \frac{m\theta + b}{100} \Rightarrow \frac{P}{P_{NOM}} \approx 1 + \frac{m + b}{100} \quad (\theta = 1) \\ \text{and} \quad \frac{P}{P_{NOM}} \bar{\theta}^{x-\alpha} &\approx 1 + \frac{m\bar{\theta} + b}{100} \quad (\theta = \bar{\theta}) \end{aligned}$$

therefore

$$\bar{\theta}^{x-\alpha} \approx \left[1 + \frac{m\bar{\theta} + b}{100} \right] / \left[1 + \frac{m + b}{100} \right]$$

Taking natural logs of both sides and applying the relationship that $\ln(1+x) \approx x$ for small x , we obtain an expression for the new exponent in terms of the old exponent, namely,

$$x \approx \alpha + \frac{m(\bar{\theta} - 1)}{100 \ln(\bar{\theta})} \quad (6)$$

We see from this equation, that if the slope of the regression line is positive, then the exponent will be raised. Likewise, it will be lowered when the slope is negative. If the data sample was such that the mean theta was standard day (i.e., =1), then note that the limit of $(\bar{\theta}-1)/\ln(\bar{\theta})$ as $\bar{\theta} \rightarrow 1$, is 1, and the equation reduces to

$$x \approx \alpha + \frac{m}{100} \quad (7)$$

The form of these equations is particularly simple and allows a rapid check for corrections when engine data is available. The reader should note that the above can be applied just as well to pressure corrections (i.e. the δ exponent) where the relationships would be identical except for replacing θ with δ .

There are, of course, other empirical paths which one can take in pursuit of the proper correction. For instance, one could determine the exponent which minimizes the standard deviation ($\sigma_{\Delta G}$) of the parameter deltas $\{\Delta G_i\}$, or to minimize the covariance ($\sigma_{\Delta G, \theta}$) between the deltas $\{\Delta G_i\}$ and $\{\theta_i\}$, or the correlation coefficient ($r_{\Delta G, \theta}$). Monte Carlo simulations indicate, however, that the results are approximately the same for all of these techniques. The differences obtained for theta exponents using these four methods are well within the limits of measurement noise and are not statistically significant. The three minimization methods which we have just alluded to, do carry a substantial computational overhead in that each of these methods are iterative in nature and require that a complete statistical analysis be performed at each newly chosen exponent, where the exponent is modulated in small increments over some range. After that is completed, the minimum statistic is isolated and the corresponding exponent is selected as the winner. In this regard, the simple formula(s) given in Eqs. (6) and (7) above are superior in that they return essentially the same exponent but require only one statistical analysis.

As an illustration, Table 4 below presents the results of a Monte Carlo simulation comparison of the slope formula method and the minimum standard deviation method. This table contains 50 randomly chosen *true* exponents between 0.7 and 1.0. For *each* exponent a set of 200 deltas $\{\Delta G_i\}$ were randomly generated. These deltas were selected from a normal population where the variance for each of the 50 $\{\Delta G_i\}$ sets was also selected randomly. Thus, we have created a set of data where we have differing *true* exponent magnitudes and as well as differing variation in the deltas (simulating different measurement accuracies and/or engine to engine variation levels). For each of these 50 true exponents sets, we set a fixed corrected parameter level P^* and generate 200 θ_i 's and calculate a set of 200 un-corrected parameters $\{P_i\}$ by applying *both* the theta correction (in reverse) using the true exponent, as well as the % delta $\{\Delta G_i\}$. This produces a set of data that simulates 200 raw measurements, each at a different temperature, each with a different (measurement/engine) bias ($\{\Delta G_i\}$). The $\{P_i\}$ are now *corrected* using one of 50 randomly chosen incorrect reduction theta exponents. Using this set of falsely corrected

parameters, the aforementioned 2 methods are applied and an estimate for the true exponent determined. This is repeated for each of the 50 samples exponents. Table 4 contains a sampling of the data generated. Figure 3 below summarizes the results of all 50 sets in graphical form.

Table 4: Monte Carlo Simulation

TRUE Exp	Reduction Exp	Slope Method		Min Std Deviation Method	
		Exp	Error	Exp	Error
0.710	0.789	0.707	-0.0035	0.705	-0.005
0.702	0.899	0.710	0.0085	0.710	0.0082
0.953	0.756	0.954	0.0008	0.950	-0.0028
0.773	0.864	0.768	-0.0046	0.775	0.0025
0.866	0.827	0.866	0.0006	0.865	-0.0005
0.964	0.710	0.986	0.0218	0.980	0.0158
0.743	0.936	0.734	-0.0086	0.735	-0.0075
0.749	0.706	0.763	0.0135	0.760	0.0109
0.953	0.701	0.950	-0.0023	0.950	-0.0025
0.708	0.809	0.711	0.0027	0.715	0.0071
0.791	0.787	0.784	-0.0072	0.785	-0.0063
0.961	0.747	0.956	-0.0045	0.960	-0.0008
0.775	0.893	0.738	-0.0366	0.740	-0.0347
0.860	0.934	0.850	-0.0098	0.855	-0.0052
.	.	.	.	.	.
.	.	.	.	.	.
0.920	0.988	0.913	-0.0079	0.915	-0.0054
0.729	0.703	0.727	-0.0017	0.725	-0.0039
0.821	0.729	0.844	0.0223	0.845	0.0237
0.867	0.972	0.862	-0.005	0.865	-0.0018
0.774	0.943	0.770	-0.0039	0.770	-0.0035
0.864	0.811	0.864	-0.0007	0.865	0.0007
0.927	0.711	0.929	0.0021	0.930	0.003
0.978	0.986	0.976	-0.0017	0.975	-0.0029
0.756	0.796	0.757	0.0004	0.755	-0.0012
Mean Error =			-0.0011		-0.00053

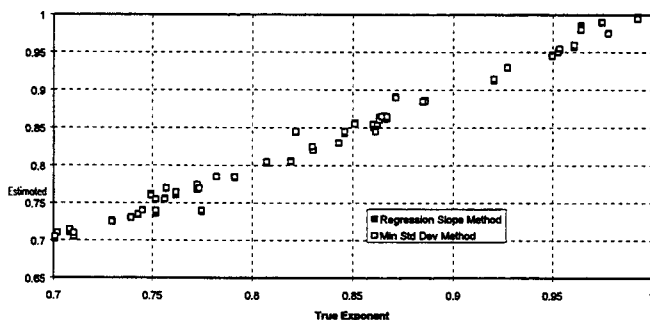


Figure 3: Monte Carlo Simulation Experiment

As can be seen by perusing the table, a reasonably accurate assessment of the proper exponent can be obtained using the empirical method described above. In addition, both methods used in this example return exponents which are essentially the same. The slope

method, however, offers the additional advantage of requiring far less computational effort

SUMMARY

We have summarized the traditional standard day corrections for most of the commonly encountered gas path parameters such as temperatures, pressures, speeds and flows as well as providing a derivation for each utilizing only simple thermodynamic relationships and fundamentals of the calculus. The topic of dimensional analysis was also mentioned as a matter of historical perspective in the treatment of non-dimensional quantities and their relationship to corrected parameters. Other model dependent analytical methods were also discussed and illustrated. Finally, and perhaps most importantly, the practical aspects of determining the theta-delta exponents empirically from samples of engine data was discussed and a computationally simple method was derived and illustrated.

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